

Total No. of Questions : 4]

SEAT No. :

PE-538

[Total No. of Pages : 2

[6578]-11

**S.E. (Electronics / E&Tc / E&C (VLSI Design & Technology)/
(Advanced Communication Technology)) (Insem.)**

ENGINEERING MATHEMATICS - III

(2019 Pattern) (Semester - III) (207005)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2 and Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of non-programmable scientific calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Solve any two :

[10]

- i) $(D^3 + 4D)y = \sin 2x.$
- ii) Solve by method of variation of parameters
 $(D^2 - 4D + 4)y = e^{2x} \cdot \sec^2 x.$
- iii) $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$

b) Solve : $\frac{dx}{y + zx} = \frac{dy}{-x - yz} = \frac{dz}{x^2 - y^2}.$

[5]

OR

Q2) a) Solve any two :

[10]

- i) $(D^2 + 6D + 9)y = \frac{1}{x^3} e^{-3x}.$
- ii) Solve by method of variation of parameters $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x.$
- iii) $(3x + 2)^2 \frac{d^2 y}{dx^2} + 3(3x + 2) \frac{dy}{dx} - 36y = 3x$

- b) An inductor of 0.5 henries is connected in series with a resistor of 6 ohms, a capacitor of 0.02 Farads, a generator having alternative voltage given by $24 \sin(10t)$, $t > 0$ At $t = 0$, the charge Q and the current I are zero. Find the charge Q at any time t .

[5]

P.T.O.

Q3) a) Find Fourier sine transform of $f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ 2-x & 1 < x \leq 2 \\ 0 & x > 2 \end{cases}$. [5]

b) Attempt any One : [5]

i) Find z-transform of $f(k) = 3^k \cos 4k$, $k \geq 0$.

ii) Find inverse z-transform of $F(z) = \frac{1}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)}$, $|z| > \frac{1}{2}$.

c) Solve the following difference equation

$$f(k+1) + \frac{1}{4} f(k) = \left(\frac{1}{4}\right)^k, k \geq 0, f(0) = 0. \quad [5]$$

OR

Q4) a) Using Fourier integral representation show that

$$\frac{2}{\pi} \int_0^\infty \frac{k \cos \lambda x}{\lambda^2 + k^2} d\lambda = e^{-kx}, k > 0. \quad [5]$$

b) Attempt any one : [5]

i) Find z-transform of $f(k) = k^2$, $k \geq 0$

ii) Find inverse z-transform of $F(z) = \frac{2z^2 - 3z}{(z+2)(z-4)}$, $|z| < 2$.

c) Solve : $\int_0^\infty f(x) \sin \lambda x dx = \begin{cases} 1, & 0 < \lambda < 1 \\ 2, & 1 < \lambda < 2 \\ 0, & \lambda > 2 \end{cases}$. [5]
